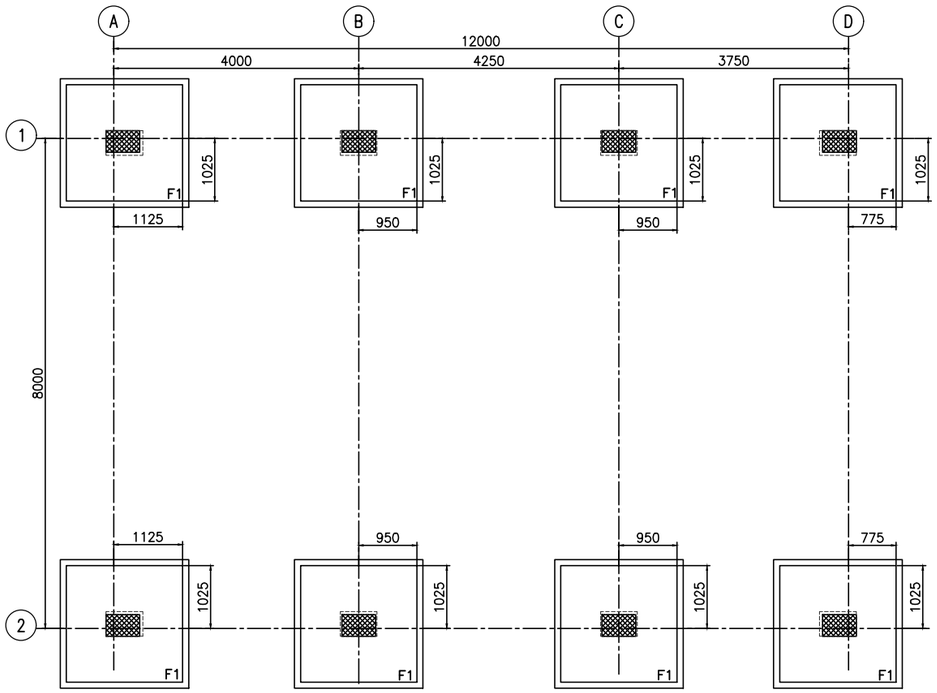
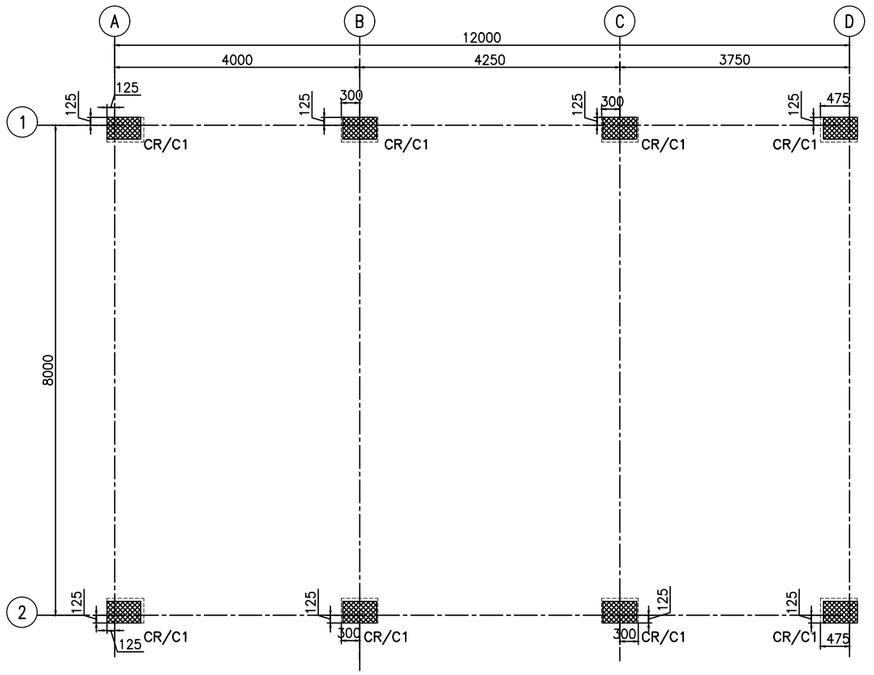
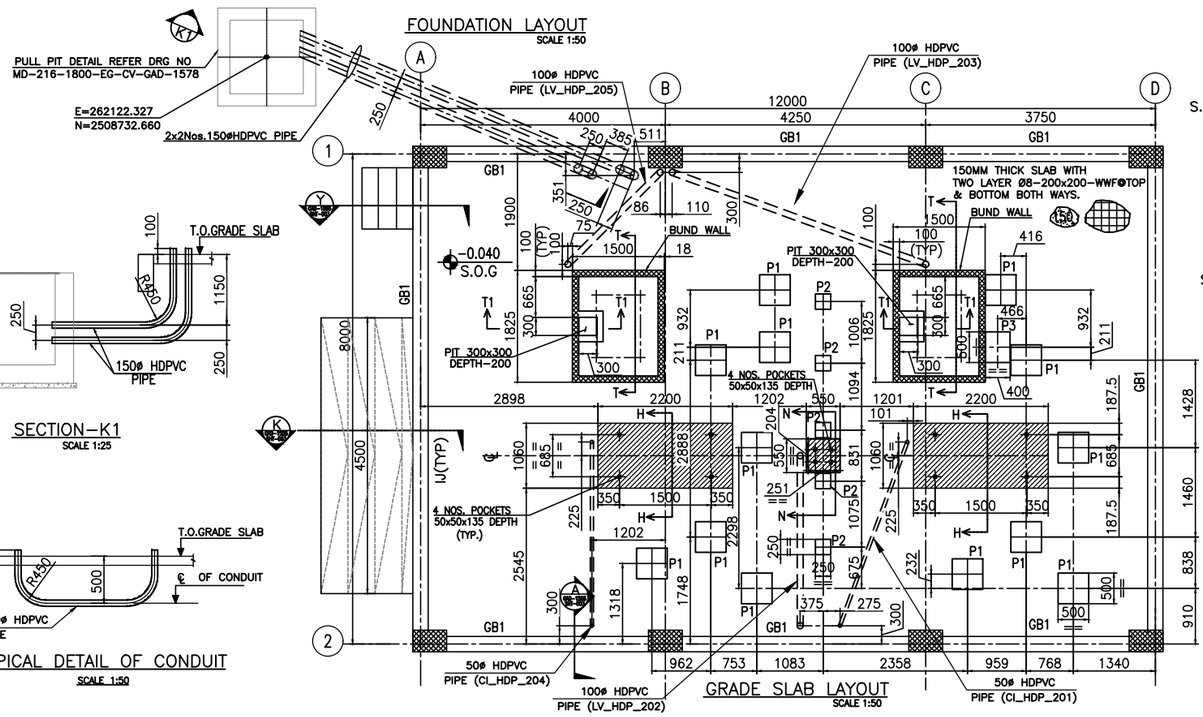
DRAWING NUMBER MD-216-1800-EG-CV-GAD-1555 REV D01



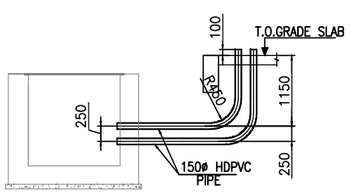
FOUNDATION LAYOUT
SCALE 1:50



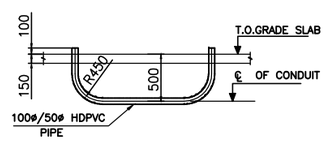
CONCRETE COLUMN LAYOUT
SCALE 1:50



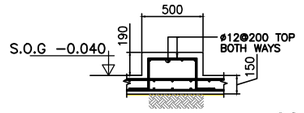
GRADE SLAB LAYOUT
SCALE 1:50



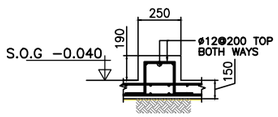
SECTION–K1
SCALE 1:25



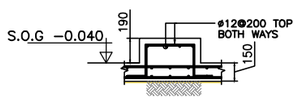
TYPICAL DETAIL OF CONDUIT
SCALE 1:50



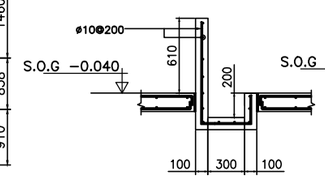
TYP. DETAIL OF P1
SCALE 1:25



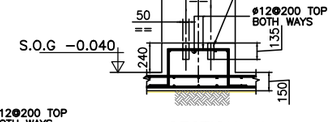
TYP. DETAIL OF P2
SCALE 1:25



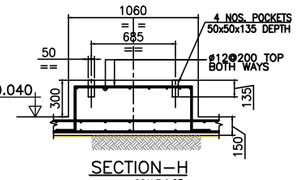
TYP. DETAIL OF P3
SCALE 1:25



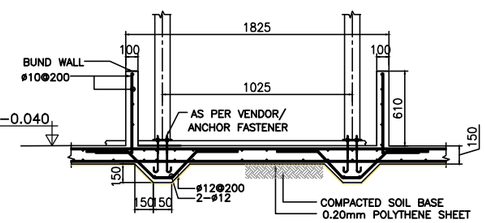
SECTION–T1
SCALE 1:25



SECTION–N
SCALE 1:25

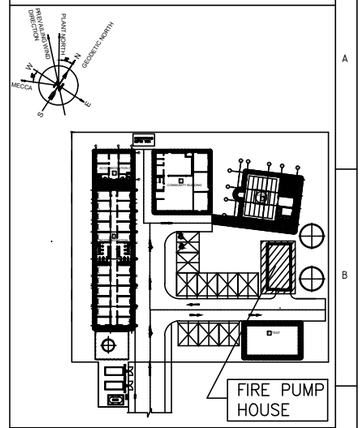


SECTION–H
SCALE 1:25

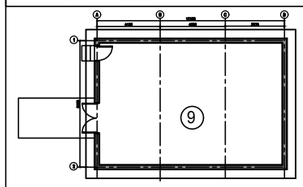


SECTION–T
SCALE 1:25

MASTER KEY PLAN



FIRE PUMP HOUSE ENLARGED KEYPLAN



FIRE PUMP HOUSE			
BLDG NO.	CORNER NO.	EASTING	NORTHING
9	A1	E 262122.062	N 2508735.588
	D2	E 262122.545	N 2508749.628

NOTES

1.) ALL DIMENSIONS ARE GIVEN IN MILLIMETRES AND ELEVATIONS ARE IN METRES UNLESS OTHERWISE NOTED.
2.) MINIMUM REINFORCED CONCRETE GRADE TO BE USED FOR SUPERSTRUCTURE & SUBSTRUCTURES IS M35(5000 PSI).
3.) MINIMUM LEAN CONCRETE GRADE TO BE USED FOR THE SUB-STRUCTURES IS M15(2175 PSI).
4.) ALL REINFORCING STEEL SHALL BE DEFORMED BILLET STEEL BARS CONFORMING TO ASTM-A615, WITH MINIMUM YIELD STRENGTH OF FY420 MPA.
5.) THE NET SOIL BEARING CAPACITY CONSIDERED IN FOUNDATION DESIGN IS 350 KPA(REFER DOC 216-3000-EG-CV-CAL-1008).
6.) WATER PROOFING INSULATION SHALL BE AS PER PROJECT SPECS AND ITS MATERIAL TO BE APPROVED BY ENGINEER ON SITE.
7.) THE MINIMUM CONCRETE COVER AS PER SBC-304-2018(TABLE 20.6.1.3.1) MEASURED FROM REBAR SHALL BE
   a) CONCRETE CAST AGAINST & PERMANENTLY EXPOSED TO EARTH (INCLUDING FOUNDATIONS OVER SUB-SLAB) FORMED CONCRETE EXPOSED TO EARTH OR WEATHER.....50MM
   b.) CONCRETE NOT EXPOSED TO WEATHER NOR IN CONTACT WITH GROUND
   BEAMS, COLUMNS.....40MM
   SLABS, WALLS, JOISTS.....20MM
   c.) CONCRETE SLAB OVER 50MM SUB-SLAB.....20MM
8.) ALL DRAWINGS MUST BE READ IN CONJUNCTION WITH RELEVANT ARCHITECTURAL AND ENGINEERING DRAWINGS RELATED TO THE SAME SHEET AND ANY ITEM ON IT.
9.) F.F.L +0.00 MENTIONED IS EQUIVALENT TO RL 923.70 AS PER THE CIVIL DRAWING CE-004A-OVERALL LAYOUT PLAN.

LEGENDS
F-FOUNDATION
C-COLUMN
FFL-FINISH FLOOR LVL
FGL-FINISH GRADE LVL
TOS-TOP OF SCREED
WWF-WELDED WIRE FABRIC
HCS-HOLLOW CORE SLAB
GB-GRADE BEAM
IJ-ISOLATION JOINT
CJ-CONTROL JOINT
GA-GALVANIZED
S.O.G-SLAB ON GRADE
ARCH-ARCHITECTURAL
HOR.-HORIZONTAL
REINF.-REINFORCEMENT
LVL.-LEVEL
RL-REDUCED LEVEL
℄-CENTRE LINE
CR-COLUMN NECK
EQUIPMENT PAD
BUND WALL
CR-COLUMN NECK

SHEET 1 OF 3

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-39LA-EG-FG-GAD-V513-2004	VILLAGE-FIRE PUMP HOUSE MISCELLANEOUS DETAIL								
MD-216-39LA-EG-FG-LAD-V513-2029	SECURITY FORCE BASE CAMP - FIRE PUMP HOUSE								
MD-216-1800-EG-EL-LAD-1015	CABLE ROUTE LAYOUT FIREWATER PUMP HOUSE								
MD-216-6000-EG-CV-STD-1002	HOLLOW CORE SLAB STANDARD	D01	27/02/2023	AS-BUILT	--	AVG	--	SJM	
MD-216-6000-EG-CV-STD-1001	REINFORCEMENT STANDARD	C02	22/06/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED △02	--	AVG	--	SJM	
MD-216-39LA-EG-FG-GAD-V513	BUILDING PLANS	C01	27/11/2020	ISSUED FOR CONSTRUCTION	--	AVG	--	SJM	
MD-216-6000-EG-CV-GAD-1535	OVERALL LAYOUT PLAN	A01	22/10/2020	ISSUED FOR APPROVAL	--	AVG	--	SJM	

DESIGNER: مكتب الخليج للاستشارات الهندسية – شركة المملكة العربية السعودية
GULF CONSULT ARCHITECTS AND ENGINEERS

DRAWING NUMBER					REVISION
SC-PH-ST-001					A01
DATE	15/10/2020	DATE	16/10/2020	DATE	17/10/2020
DRAWN	MAH	CHECKED	MF	APPROVED	FK

LARSEN & TOUBRO DRAWING NUMBER: 019045_900_C_G_24256_001 REVISION 3
OUTOTEC DRAWING NUMBER: 900707__GAU01__AN02__MTL01__18167 DOCUMENT ID: OU100696616 REVISION 3

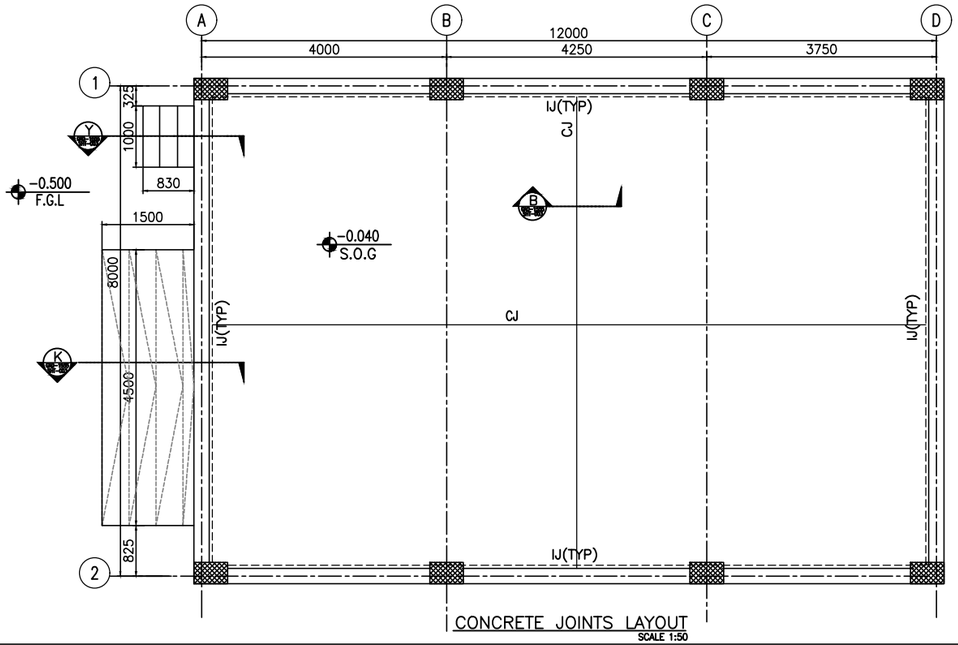
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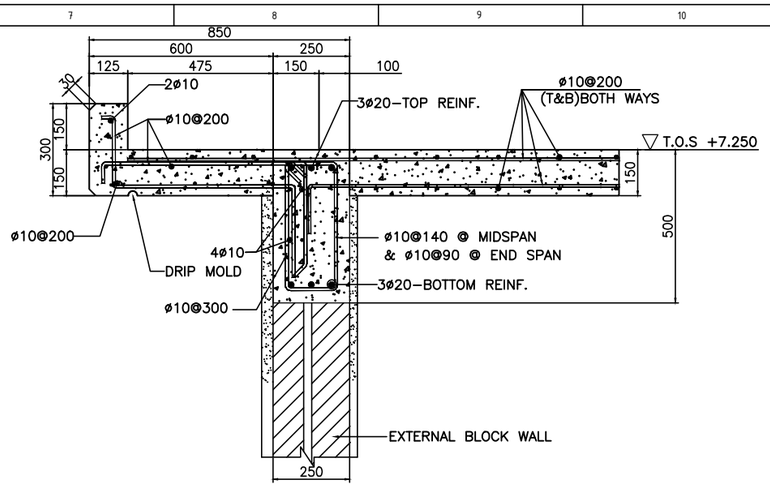
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DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	AVG	21/10/2020	GENERAL ARRANGEMENT
ENGINEER	--	--	SECURITY BASE CAMP:FIRE FIGHTING PUMP HOUSE
APPROVED	SJM	22/10/2020	LAYOUT OF FDN, COL & GRADE BEAM
EMPLOYER			
SCALE	1:50	A1	DRAWING NUMBER: MD-216-1800-EG-CV-GAD-1555 REV: D01

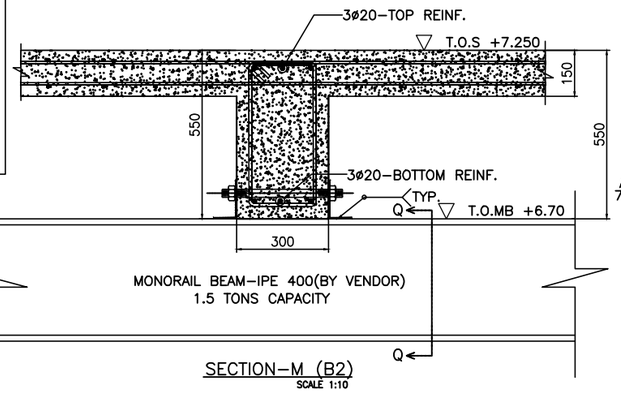
DRAWING NUMBER MD-216-1800-EG-CV-GAD-1555 REV D01



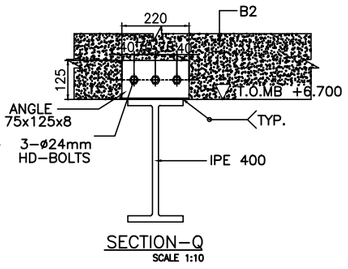
CONCRETE JOINTS LAYOUT
SCALE 1:50



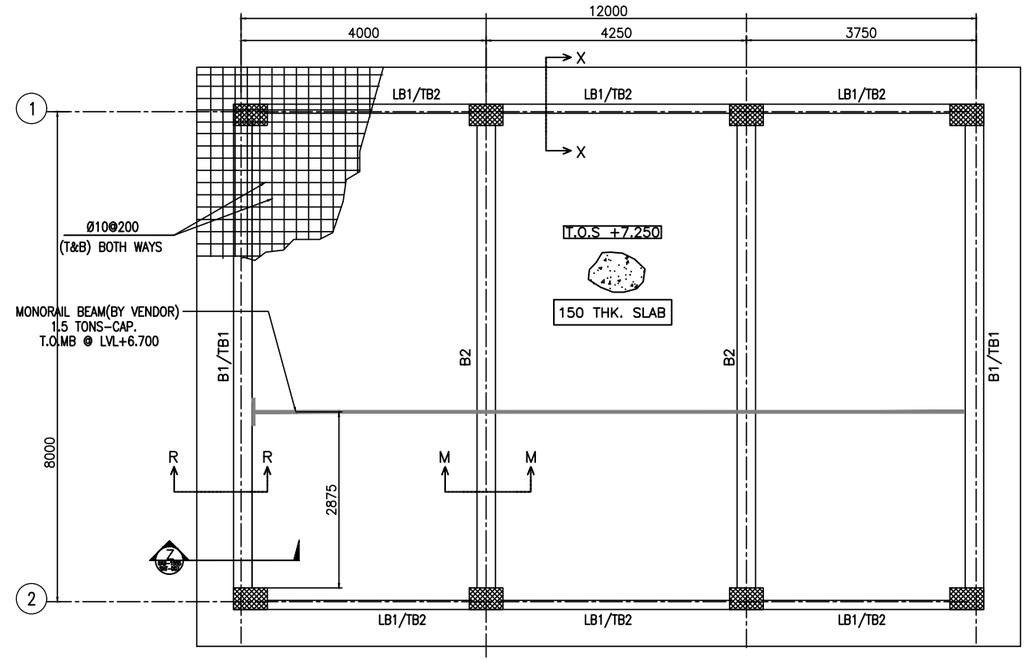
SECTION-X (LB1)
SCALE 1:10



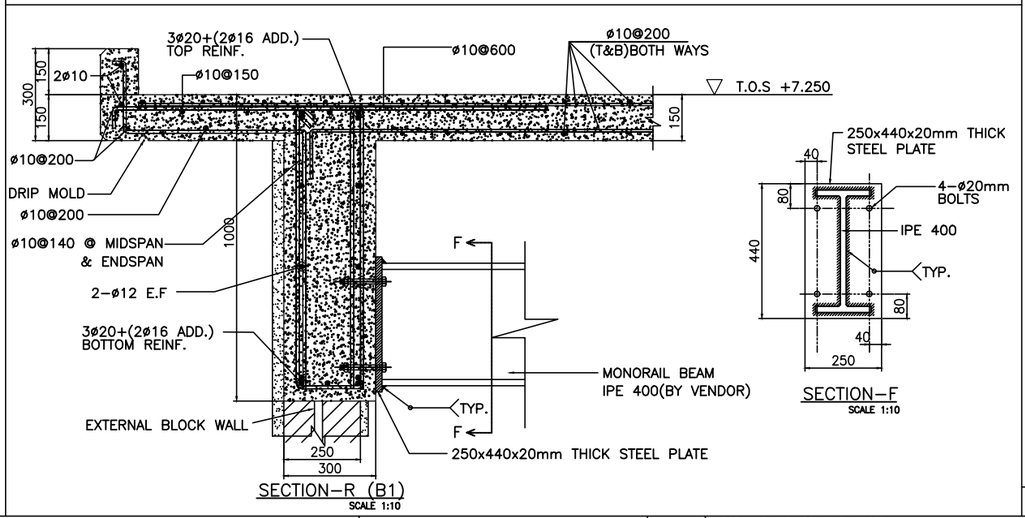
SECTION-M (B2)
SCALE 1:10



SECTION-Q
SCALE 1:10



ROOF SLAB LAYOUT
SCALE 1:50



SECTION-R (B1)
SCALE 1:10

SECTION-F
SCALE 1:10

NOTES

1.) ALL DIMENSIONS ARE GIVEN IN MILLIMETRES AND ELEVATIONS ARE IN METRES UNLESS OTHERWISE NOTED.
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3.) MINIMUM LEAN CONCRETE GRADE TO BE USED FOR THE SUB-STRUCTURES IS M15(2175 PSI).
4.) ALL REINFORCING STEEL SHALL BE DEFORMED BILLET STEEL BARS CONFORMING TO ASTM-A615, WITH MINIMUM YIELD STRENGTH OF FY420 MPA .
5.) THE NET SOIL BEARING CAPACITY CONSIDERED IN FOUNDATION DESIGN IS 350 KPA(REFER DOC 216-3000-EG-CV-CAL-1008).
6.) WATER PROOFING INSULATION SHALL BE AS PER PROJECT SPECS AND ITS MATERIAL TO BE APPROVED BY ENGINEER ON SITE.
7.) THE MINIMUM CONCRETE COVER AS PER SBC-304-2018(TABLE 20.6.1.3.1) MEASURED FROM REBAR SHALL BE
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   b.) CONCRETE NOT EXPOSED TO WEATHER NOR IN CONTACT WITH GROUND
   BEAMS, COLUMNS.....40MM
   SLABS, WALLS, JOISTS.....20MM
   c.) CONCRETE SLAB OVER 50MM SUB-SLAB.....20MM
8.) ALL DRAWINGS MUST BE READ IN CONJUNCTION WITH RELEVANT ARCHITECTURAL AND ENGINEERING DRAWINGS RELATED TO THE SAME SHEET AND ANY ITEM ON IT.
9.) F.F.L +0.00 MENTIONED IS EQUIVALENT TO RL RL 923.70 AS PER THE CIVIL DRAWING CE-004A-OVERALL LAYOUT PLAN.

LEGENDS
F-FOUNDATION
C-COLUMN
FFL-FINISH FLOOR LVL
FGL-FINISH GRADE LVL
TOS-TOP OF SCREED
WWF-WELDED WIRE FABRIC
HCS-HOLLOW CORE SLAB
GB-GRADE BEAM
IJ-ISOLATION JOINT
CJ-CONTROL JOINT
SCJ-SLAB CONSTRUCTION JOINT
GA-GALVANIZED
S.O.G-SLAB ON GRADE
ARCH-ARCHITECTURAL
HOR.-HORIZONTAL
REINF.-REINFORCEMENT
LVL.-LEVEL
TOS-TOP OF SCREED
RL-REDUCED LEVEL
TB-TIE BEAM/BOND BEAM
B/LB1-ROOF BEAMS
TOMB-TOP OF MONORAIL BEAM
CAP.-CAPACITY
E.F-EACH FACE
(T&B)-TOP & BOTTOM

SHEET 2 OF 3

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-6000-EG-CV-STD-1002	HOLLOW CORE SLAB STANDARD	D01	27/02/2023	AS-BUILT	--	AVG	--	SJM	
MD-216-6000-EG-CV-STD-1001	REINFORCEMENT STANDARD	C02	22/06/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED (02)	--	AVG	--	SJM	
MD-216-39LA-EG-FG-GAD-V513	BUILDING PLANS	C01	27/11/2020	ISSUED FOR CONSTRUCTION	--	AVG	--	SJM	
MD-216-6000-EG-CV-GAD-1535	OVERALL LAYOUT PLAN	A01	22/10/2020	ISSUED FOR APPROVAL	--	AVG	--	SJM	

DESIGNER: GULF CONSULT ARCHITECTS AND ENGINEERS

DRAWING NUMBER	REVISION
SC-PH-ST-002	A01

DATE	15/10/2020	DATE	16/10/2020	DATE	17/10/2020
DRAWN	MAH	CHECKED	MF	APPROVED	FK

LARSEN & TOUBRO DRAWING NUMBER: 019045_900_C_G_24256_001 — REVISION 3
OUTOTEC DRAWING NUMBER: 900707__GAU01__AN02__MTL01__18167 — DOCUMENT ID: OU100696616 — REVISION 3

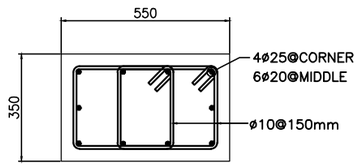
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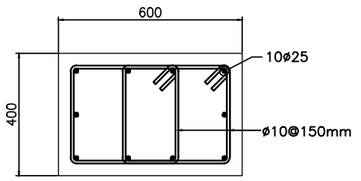
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DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	AVG	21/10/2020	GENERAL ARRANGEMENT
ENGINEER	--	--	SECURITY BASE CAMP:FIRE FIGHTING PUMP HOUSE
APPROVED	SJM	22/10/2020	LAYOUT & DETAILS OF ROOF & GRADE SLAB
EMPLOYER			
SCALE 1:50		A1	DRAWING NUMBER: MD-216-1800-EG-CV-GAD-1555 — REV: D01

DRAWING NUMBER MD-216-1800-EG-CV-GAD-1555 REV D01

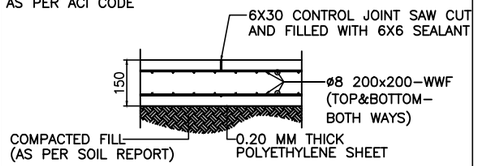


COLUMN DETAIL C1 SCALE 1:10 GAD-1555 SHT-01

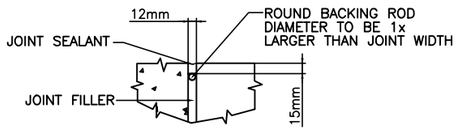


COLUMN DETAIL CR SCALE 1:10 GAD-1555 SHT-01

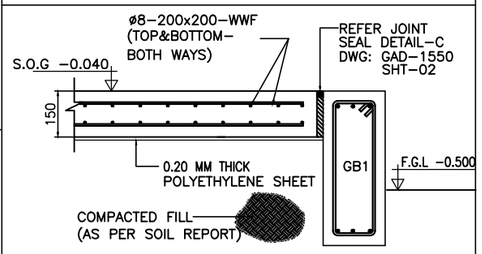
NOTE: CONTROL JOINTS SHALL BE PROVIDED AT MAXIMUM SPACING OF 20 FT (6.10 M) AS PER ACI CODE



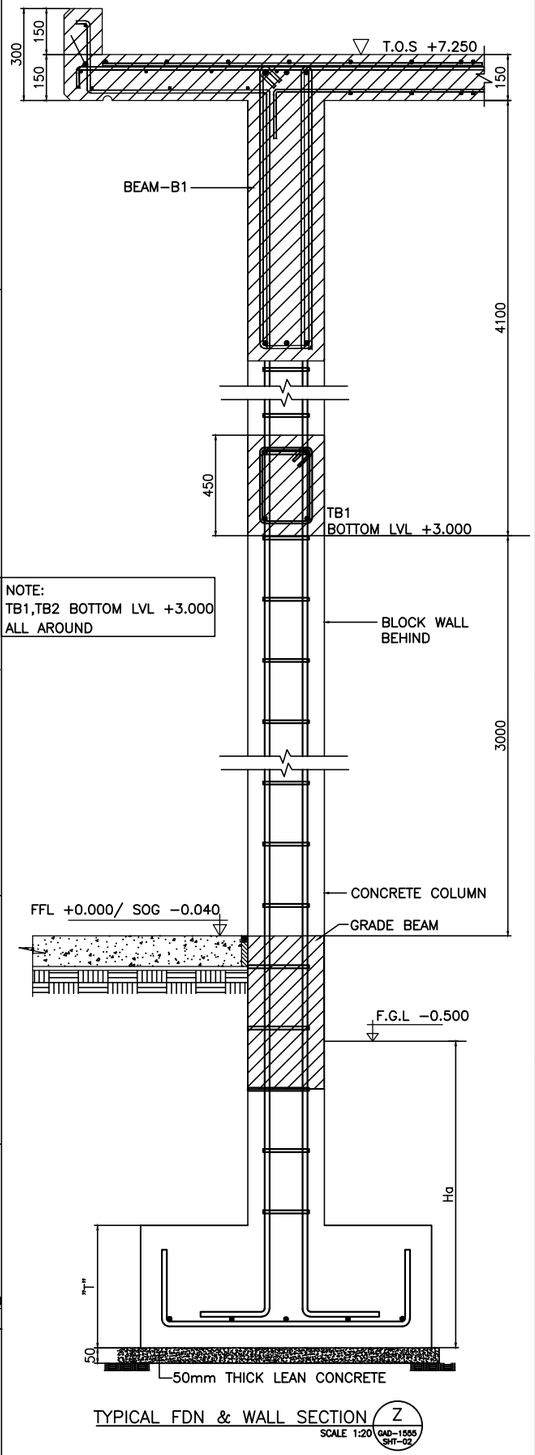
SECTION-B CJ SCALE NTS GAD-1555 SHT-02



TYP JOINT SEAL DETAIL C SCALE NTS GAD-1555 SHT-02



TYP SECTION FOR GRADE SLAB A SCALE 1:10 GAD-1555 SHT-01



NOTE:
TB1,TB2 BOTTOM LVL +3.000 ALL AROUND

TYPICAL FDN & WALL SECTION Z SCALE 1:20 GAD-1555 SHT-02

SCHEDULE OF FOOTINGS

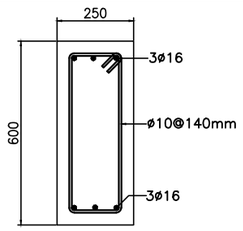
FOOTING MARK	PLAIN CONCRETE			REINFORCED CONCRETE			BOTTOM REINFORCEMENT		TOP REINFORCEMENT		Ha	REMARKS
	L	W	T	L	W	T	SHORT BARS	LONG BARS	SHORT BARS	LONG BARS		
F1	2100	2100	50	1900	1900	450	ø12@125	ø12@125	–	–	1500	

SCHEDULE OF GRADE BEAMS

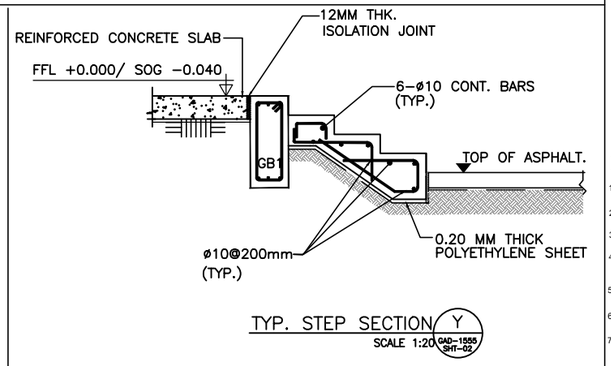
BEAM MARK	DIMENSIONS		STEEL REINFORCEMENT				STIRRUPS
			BOTTOM		TOP		
	WIDTH	DEPTH	STRAIGHT	ADD.	STRAIGHT	ADD.	
GB1	250	600	3ø16	–	3ø16	–	PROVIDE AT THE ⊕DISTANCE OF L/3 FROM THE SUPPORT IN ø10@140 IN THE END SPAN & MID SPAN

SCHEDULE OF ROOF BEAMS

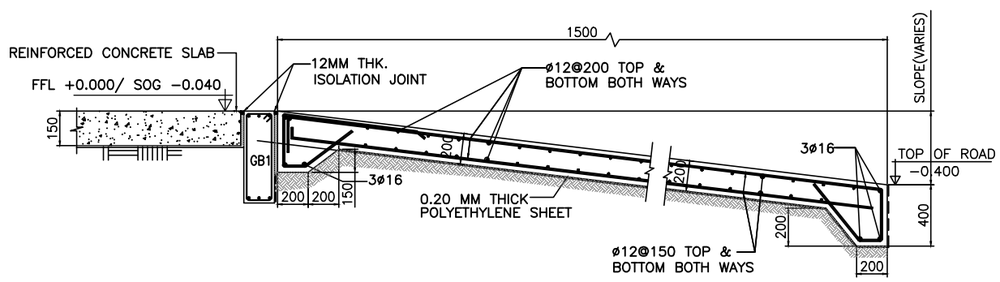
BEAM MARK	DIMENSIONS		STEEL REINFORCEMENT					STIRRUPS	REMARKS
			TOP		BOTTOM		SIDE BAR		
	WIDTH	DEPTH	STRAIGHT	ADD.	STRAIGHT	ADD.	EACH FACE		
LB1	250	500	3ø20	–	3ø20	–	–	PROVIDE ø10⊕90 AT THE ⊕DISTANCE OF L/3 FROM THE SUPPORT IN END SPAN & ø10@140 IN THE MID SPAN	REFER DETAIL-X
B1	300	1000	3ø20	2ø16	3ø20	2ø16	2ø12	PROVIDE ø10@140 AT MID SPAN & END SPAN	REFER DETAIL-R
B2	300	550	3ø20	–	3ø20	–	–	PROVIDE ø10⊕90 AT THE ⊕DISTANCE OF L/3 FROM THE SUPPORT IN END SPAN & ø10@140 IN THE MID SPAN	REFER DETAIL-M
TB1	250	450	3ø12	–	3ø12	–	–	PROVIDE ø10⊕200 AT MID SPAN & END SPAN	BOTTOM⊕LVL 3.000
TB2	250	250	2ø12	–	2ø12	–	–	PROVIDE ø10⊕200 AT MID SPAN & END SPAN	BOTTOM⊕LVL 3.000



GB1 @ MIDSPAN & ENDSPAN S SCALE 1:20 GAD-1555 SHT-01



TYP. STEP SECTION Y SCALE 1:20 GAD-1555 SHT-02



TYPICAL RAMP SECTION K SCALE 1:20 GAD-1555 SHT-02

NOTES

1.) ALL DIMENSIONS ARE GIVEN IN MILLIMETRES AND ELEVATIONS ARE IN METRES UNLESS OTHERWISE NOTED.
2.) MINIMUM REINFORCED CONCRETE GRADE TO BE USED FOR SUPERSTRUCTURE & SUBSTRUCTURES IS M35(5000 PSI).
3.) MINIMUM LEAN CONCRETE GRADE TO BE USED FOR THE SUB-STRUCTURES IS M15(2175 PSI).
4.) ALL REINFORCING STEEL SHALL BE DEFORMED BILLET STEEL BARS CONFORMING TO ASTM-A615, WITH MINIMUM YIELD STRENGTH OF FY420 MPA.
5.) THE NET SOIL BEARING CAPACITY CONSIDERED IN FOUNDATION DESIGN IS 350 KPA(REFER DOC 216-3000-EG-CV-CAL-1008).
6.) WATER PROOFING INSULATION SHALL BE AS PER PROJECT SPECS AND ITS MATERIAL TO BE APPROVED BY ENGINEER ON SITE.
7.) THE MINIMUM CONCRETE COVER AS PER SBC-304-2018(TABLE 20.6.1.3.1) MEASURED FROM REBAR SHALL BE
   a) CONCRETE CAST AGAINST & PERMANENTLY EXPOSED TO EARTH (INCLUDING FOUNDATIONS OVER SUB-SLAB) FORMED CONCRETE EXPOSED TO EARTH OR WEATHER.....50MM
   b.) CONCRETE NOT EXPOSED TO WEATHER NOR IN CONTACT WITH GROUND
   BEAMS, COLUMNS.....40MM
   SLABS, WALLS, JOISTS.....20MM
   c.) CONCRETE SLAB OVER 50MM SUB-SLAB.....20MM
8.) ALL DRAWINGS MUST BE READ IN CONJUNCTION WITH RELEVANT ARCHITECTURAL AND ENGINEERING DRAWINGS RELATED TO THE SAME SHEET AND ANY ITEM ON IT.
9.) F.F.L +0.00 MENTIONED IS EQUIVALENT TO RL 923.70 AS PER THE CIVIL DRAWING CE-004A-OVERALL LAYOUT PLAN.

LEGENDS
F-FOUNDATION
C-COLUMN
FFL-FINISH FLOOR LVL
FGL-FINISH GRADE LVL
TOS-TOP OF SCREED
WWF-WELDED WIRE FABRIC
HCS-HOLLOW CORE SLAB
GB-GRADE BEAM
IJ-ISOLATION JOINT
CJ-CONTROL JOINT
GA-GALVANIZED
S.O.G-SLAB ON GRADE
ARCH-ARCHITECTURAL
HOR.-HORIZONTAL
REINF.-REINFORCEMENT
LVL-LEVEL
RL-REDUCED LEVEL
VER-VERTICAL
FDN-FOUNDATION
TB-TIE BEAM/BOND BEAM
B/LB1-ROOF BEAMS

SHEET 3 OF 3

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-6000-EG-CV-STD-1002	HOLLOW CORE SLAB STANDARD	D01	27/02/2023	AS-BUILT	--	AVG	--	SJM	
MD-216-6000-EG-CV-STD-1001	REINFORCEMENT STANDARD	C02	22/06/2021	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED △02	--	AVG	--	SJM	
MD-216-39LA-EG-FG-GAD-V513	BUILDING PLANS	C01	27/11/2020	ISSUED FOR CONSTRUCTION	--	AVG	--	SJM	
MD-216-6000-EG-CV-GAD-1535	OVERALL LAYOUT PLAN	A01	22/10/2020	ISSUED FOR APPROVAL	--	AVG	--	SJM	

DESIGNER: مكتب الخليج للاستشارات الهندسية - الغير - الشركة العربية السعودية
GULF CONSULT ARCHITECTS AND ENGINEERS

DRAWING NUMBER						REVISION
SC-PH-ST-003						A01
DATE	15/10/2020	DATE	16/10/2020	DATE	17/10/2020	
DRAWN	MAH	CHECKED	MF	APPROVED	FK	

LARSEN & TOUBRO DRAWING NUMBER	REVISION
019045_900_C_G_24256_001	3
OUTOTEC DRAWING NUMBER	DOCUMENT ID / REVISION
900707__GAU01__AN02__MTL01__18167	OU100696616 / 3

LARSEN & TOUBRO

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DRAWN	--	--
CHECKED	AVG	21/10/2020
ENGINEER	--	--
APPROVED	SJM	22/10/2020
EMPLOYER		
SCALE	1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
SECURITY BASE CAMP:FIRE FIGHTING PUMP HOUSE
SECTIONS & DETAILS OF BUILDING

DRAWING NUMBER MD-216-1800-EG-CV-GAD-1555 REV D01